

YUE YANG

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EDUCATION

Ph.D. in Computer and Information Science University of Pennsylvania Thesis: Language Priors for Visual Intelligence	2020 - 2025
M.S. in Robotics University of Pennsylvania	2018 - 2020
B.E. in Mechanical Engineering Zhejiang University	2014 - 2018

PROFESSIONAL EXPERIENCE

Thinking Machines Lab , San Francisco, CA Member of Technical Staff	2026 - Present
Allen Institute for AI , Seattle, WA Research Scientist (Molmo , Molmo-2 , MolmoPoint , MolmoWeb)	2025 - 2026
University of Pennsylvania , Philadelphia, PA Research Assistant (KnoBo , LaBo , Holodeck , CoSyn)	2020 - 2025

RESEARCH INTERESTS

My research lies at the intersection of Natural Language Processing and Computer Vision. I aim to build multimodal intelligence that can **interact** with both the **digital** [II] and **physical** worlds [4]. I care about the **trustworthiness**, **robustness**, and **data efficiency** of AI systems.

SELECTED WORKS

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- [1] **Yue Yang***, Ajay Patel*, Matt Deitke, Tanmay Gupta, Luca Weihs, Andrew Head, Mark Yatskar, Chris Callison-Burch, Ranjay Krishna, Aniruddha Kembhavi, Christopher Clark. [Scaling Text-Rich Image Understanding via Code-Guided Synthetic Multimodal Data Generation](#). (**ACL 2025**, **SAC Highlights**)
 - [2] Matt Deitke*, Christopher Clark*, Sangho Lee, Rohun Tripathi, **Yue Yang**, Jae Sung Park, et al. (51 authors in total) [Molmo and PixMo: Open Weights and Open Data for State-of-the-Art Multimodal Models](#). (**CVPR 2025**, **Best Paper Honorable Mention**)
 - [3] **Yue Yang**, Mona Gandhi, Yufei Wang, Yifan Wu, Michael S. Yao, James C. Gee, Chris Callison-Burch, Mark Yatskar. [A Textbook Remedy for Domain Shifts: Knowledge Priors for Medical Image Analysis](#). (**NeurIPS 2024**, **spotlight**)
 - [4] **Yue Yang***, Fan-Yun Sun*, Luca Weihs*, Eli Vanderbilt, Alvaro Herrasti, Winson Han, Jiajun Wu, Nick Haber, Ranjay Krishna, Lingjie Liu, Chris Callison-Burch, Mark Yatskar, Aniruddha Kembhavi, Christopher Clark. [HOLODECK: Language Guided Generation of 3D Embodied AI Environments](#). (**CVPR 2024**)
 - [5] **Yue Yang**, Artemis Panagopoulou, Shenghao Zhou, Daniel Jin, Chris Callison-Burch, Mark Yatskar. [Language in a Bottle: Language Model Guided Concept Bottlenecks for Interpretable Image Classification](#). (**CVPR 2023**)

PUBLICATIONS

1. Christopher Clark, Jieyu Zhang, Zixian Ma, Jae Sung Park, Mohammadreza Salehi, Rohun Tripathi, Sangho Lee, Zhongzheng Ren, Chris Dongjoo Kim, Yinuo Yang, Vincent Shao, **Yue Yang**, Weikai Huang, Ziqi Gao, Taira Anderson, Jianrui Zhang, Jitesh Jain, George Stoica, Winson Han, Ali Farhadi, Ranjay Krishna. [Open Weights and Data for Vision-Language Models with Video Understanding and Grounding](#). (**CVPR** 2026, **Oral**)
2. Zhaowei Wang, Hongming Zhang, Tianqing Fang, Ye Tian, **Yue Yang**, Kaixin Ma, Xiaoman Pan, Yangqiu Song, Dong Yu. [DivScene: Evaluating Large Vision Language Models for Object Navigation with Open-Vocabulary Targets in Diverse Scenes](#). (Findings of **EMNLP** 2025)
3. Zhantao Yang, Ruili Feng, Keyu Yan, Huangji Wang, Zhicai Wang, Shangwen Zhu, Han Zhang, Jie Xiao, Pingyu Wu, Kai Zhu, Jixuan Chen, Chen-Wei Xie, **Yue Yang**, Hongyang Zhang, Yu Liu, Fan Cheng. [BACON: Improving Clarity of Image Captions via Bag-of-Concept Graphs](#). (**CVPR** 2025)
4. Long Le, Jason Xie, William Liang, Hung-Ju Wang, **Yue Yang**, Yecheng Jason Ma, Kyle Vedder, Arjun Krishna, Dinesh Jayaraman, Eric Eaton. [Articulate-Anything: Automatic Modeling of Articulated Objects via a Vision-Language Foundation Model](#). (**ICLR** 2025)
5. Yifan Wu, Yang Liu, **Yue Yang**, Michael S. Yao, Wenli Yang, Xuehui Shi, Lihong Yang, Dongjun Li, Yueming Liu, James C. Gee, Xuan Yang, Wen-bin Wei, Shi Gu. [A Concept-based Interpretable Model for the Diagnosis of Choroid Neoplasias using Multimodal Data](#). (**Nature Communications** 2025)
6. Runsheng Huang, Liam Dugan, **Yue Yang**, Chris Callison-Burch. [MiRAGeNews: Multi-modal Realistic AI-Generated News Detection](#). (Findings of **EMNLP** 2024)
7. Yiming Huang, Weilin Wan, **Yue Yang**, Chris Callison-Burch, Mark Yatskar, Lingjie Liu. [CoMo: Controllable Motion Generation through Language Guided Pose Code Editing](#). (**ECCV** 2024)
8. Tuhin Chakrabarty, Arkady Saakyan, Olivia Winn, Artemis Panagopoulou, **Yue Yang**, Marianna Apidianaki, Smaranda Muresan. [Large Language Models and Diffusion Models Co-Create Visual Metaphors](#). (Findings of **ACL** 2023)
9. Li Zhang, Hainiu Xu, **Yue Yang**, Shuyan Zhou, Weiqiu You, Manni Arora, Chris Callison-Burch. [Causal Reasoning About Entities and Events in Procedural Texts](#). (Findings of **EACL** 2023)
10. **Yue Yang**, Wenlin Yao, Hongming Zhang, Xiaoyang Wang, Dong Yu, Jianshu Chen. [Z-LaVI: Zero-Shot Language Solver Fueled by Visual Imagination](#). (**EMNLP** 2022)
11. **Yue Yang***, Artemis Panagopoulou*, Marianna Apidianaki, Mark Yatskar and Chris Callison-Burch. [Visualizing the Obvious: A Concreteness-based Ensemble Model for Noun Property Prediction](#). (Findings of **EMNLP** 2022)
12. Shuyan Zhou*, Li Zhang*, **Yue Yang**, Qing Lyu, Graham Neubig, Chris Callison-Burch. [Show Me More Details: Discovering Event Hierarchies from WikiHow](#). (**ACL** 2022)
13. **Yue Yang**, Artemis Panagopoulou, Qing Lyu, Li Zhang, Mark Yatskar, Chris Callison-Burch. [Visual Goal-Step Inference using wikiHow](#). (**EMNLP** 2021)

PREPRINTS & WORKSHOP PAPERS

- [I] Christopher Clark*, **Yue Yang***, Jae Sung Park*, Zixian Ma, Jieyu Zhang, Mohammadreza Salehi, Rohun Tripathi, Sangho Lee, Ranjay Krishna. [Molmo-Point: Better Pointing for VLMs with Grounding Tokens](#). (arxiv 2026)

- [II] Tanmay Gupta*, Piper Wolters*, Zixian Ma*, Peter Sushko*, Rock Yuren Pang*, Diego Llanes*, **Yue Yang***, Taira Anderson, Boyuan Zheng, Zhongzheng Ren, Harsh Trivedi, Taylor Blanton, Caleb Ouellette, Winson Han, Ali Farhadi, Ranjay Krishna. [MolmoWeb: Open Multimodal Action Policy for the Open Web](#). (arxiv 2026).
- [III] Jianrui Zhang, **Yue Yang**, Rohun Tripathi, Winson Han, Ranjay Krishna, Christopher Clark, Yong Jae Lee, Sangho Lee. [Unified Spatial-Temporal Token Scoring for Efficient Video-Language Models](#). (arxiv 2026)
- [IV] Zixuan Bian*, Ruohan Ren*, **Yue Yang**, Chris Callison-Burch. [HOLODECK 2.0: Vision-Language-Guided 3D World Generation with Editing](#). (arxiv 2025)
- [V] Josh Magnus Ludan, Qing Lyu, **Yue Yang**, Liam Dugan, Mark Yatskar, Chris Callison-Burch. [Interpretable-by-Design Text Classification with Iteratively Generated Concept Bottleneck](#). (arxiv 2023)
- [VI] **Yue Yang**, Joongwon Kim, Artemis Panagopoulou, Mark Yatskar, Chris Callison-Burch. [Induce, Edit, Retrieve: Language Grounded Multimodal Schema for Instructional Video Retrieval](#). (O-DRUM Workshop at CVPR 2022)

TEACHING

Teaching Assistant

CIS-521 Artificial Intelligence	2019 - 2022
CIS-530 Computational Linguistics	2021

ACADEMIC SERVICES

Paper Review

Computer Vision: CVPR, ECCV, ICCV, SIGGRAPH Asia.
 Natural Language Processing: ACL, EMNLP, NAACL, EACL, COLM.
 Machine Learning: NeurIPS, ICLR, ICML, TMLR.

PRESS COVERAGES

VentureBeat & Penn Engineering & MarkTechPost	2025
AI Vision, Reinvented: The Power of Synthetic Data	
Penn Engineering & Penn Today & Medical News Bulletin & Medical Xpress	2024
Training Medical AI with Knowledge, Not Shortcuts.	
TechCrunch & WIRED & MIT Technology & VentureBeat	2024
Ai2's Molmo shows open source can meet, and beat, closed multimodal models.	
Penn Engineering & Tech Xplore & Tech Times & Tech Briefs	2024
Penn Engineers Recreate Star Trek's Holodeck Using ChatGPT and Video Game Assets.	

AWARDS

SAC Highlights Award , ACL	2025
AWS-ASSET Fellowship , Amazon	2025
Best Paper Honorable Mention Award , CVPR	2025
Outstanding Intern of the Year Award , Ai2	2023
Outstanding Teaching Award , University of Pennsylvania	2020